

JACOB N PICKOS

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Research and Career Interests: *Embedded Systems, Computer Architecture, Operating Systems, Edge AI*

Education

University of Michigan

Ann Arbor, MI

Computer Engineering BSE, 3.65 GPA

August, 2024 - May, 2027

- **Selected Coursework:** Embedded System Design, Digital Logic Design, Computer Organization, Electronic Circuits, Web Systems, Data Structures and Algorithms, Discrete Mathematics, Principles of Optics, Intro to Nuclear Engineering
- **Awards:** Michigan Engineering Continuing Student Scholarship

Nashville State Community College

Nashville, TN

Computer Science AS, Economics AA, 4.00 GPA

January, 2022 - May, 2024

- **Selected Coursework:** Calc-based Physics I & II, General Chemistry I & II, Calculus I - III, Linear Algebra, Computer Science I & II, Statistics
- **Awards:** Benjamin A. Gilman International Scholarship, Mathematics Academic Achievement Award, Physics Galileo Award, Academic Service Scholarship, S.O.A.R. Student Leadership Distinction

Research Experience

Incoming at Oak Ridge National Laboratory

Oak Ridge, TN

Summer Research Intern

May 2026 - July 2026

- Extracting insights from unstructured DoE power grid reports using local LLMs with Ollama and Codex.

University of Tennessee — NSF REU program

Knoxville, TN

Summer Research Intern

May 2024 - August 2024

- Trained a Deep Q-Network from scratch and implemented it into a reinforcement learning agent.
- Developed autonomous vehicle simulations in SUMO to imitate traffic patterns in the presence of emergency vehicles.
- Coded rule-based vehicle movements in MATLAB to control the SUMO simulations.
- Created and presented a research poster titled "Making Way: Using Vehicle-Vehicle Communication and A.I. to Clear a Path for an Ambulance through Heavy Traffic."

Fisk University — NSF CREST (BioSS) program

Nashville, TN

Summer Research Intern

June 2023 - August 2023

- Applied crystallography to further research on superoxide dismutase, a protein found in *S. aureus*.
- Cultured *E. coli* for protein expression, ran nickel columns for protein purification, and tested protein purity using gel electrophoresis. Grew protein crystals via the vapor diffusion method.
- Processed data from .pdb files for use in PyMOL molecular modeling software to visualize protein structures.
- Created scientific presentations and shared findings before a panel of university faculty.

Project Experience

Multidisciplinary Design Program (MDP) — Walbridge

Jan 2026 - Present

- Built a multi-agent verification pipeline between Microsoft Copilot Studio, Power Automate, SharePoint and Excel to judge the quality and grounding of LLM responses after RAG on company data.
- Led an 8-person undergraduate team in an engineering Design Review before a panel of industry professionals.

Michigan Aeronautical Science Association — Apogee Detection Project Lead

Sept 2024 - Sept 2025

- Created `baro_sim`, a Python library with a C++ back end which gives users access to realistic barometers in software, for use in rocket flight simulations.
- Wrote high-quality C++ code for simulating a multi-stage apogee detection system, integrating data from barometric and IMU sensors.
- Operated within a large team-based firmware repository on GitHub, applying best-practice version control.
- Led a Preliminary Design Review for the apogee detection system, inviting contributions from multiple engineers across our team.

EECS Class Highlights

- EECS 373 - Engineered major aspects of a fully functional vending machine, including a PS/2 magstripe reader (via GPIO interrupts), stepper motors (via PWM), and circuits that delivered power to 12 individual components.

- EECS 485 - Built a full-stack web application with a JavaScript client-side frontend, Flask REST API backend, SQLite database, and authenticated user sessions, deployed on AWS EC2.
- EECS 370 - Programmed a complete ISA toolchain in C: assembler, linker, ISA simulator, cache simulator, and a multi-pipelined processor with hazard detection and forwarding.
- EECS 270 - Applied RTL design principles to implement a sequential four-function calculator on an FPGA that responds to button presses and displays output on an LED HEX display.
- EECS 215 - Integrated deep understanding of electrical circuits with engineering aptitude to build, analyze, and debug complex filters and amplifier circuits in lab.

Technical Skills

Languages: C++, C, MATLAB, Verilog, Assembly (ARMv8 subset), Python, HTML, LaTeX

Developer Tools: VS Code, Git, Unix shell, Flask, React, REST APIs, STM32CubeIDE

Technologies: LTspice, ModelSim, MATLAB's Deep Learning Toolbox, SUMO

Work Experience

Nashville State Community College

Nashville, TN

Academic Tutor

Oct 2022 - April 2023

- Assisted students in C++ programming, mathematics (general math through Calc II), and writing.

Uber/Lyft

Nashville, TN

Rideshare Driver

Oct 2020 - Jan 2024

- Logged over 2,000 rides in the Nashville area while maintaining a perfect 5-star rating across both Uber and Lyft.

Two Men and a Truck

Nashville, TN

Crew Leader/Driver

May 2020 - Oct 2020

- Led crews of professional movers on local and cross-country moves.
- Operated commercial vehicles over state lines while complying with DOT regulations.

L&L Marketplace LLC

Nashville, TN

Property Maintenance Tech/Forklift Operator

June 2019 - May 2020

- Aided in the construction and ongoing maintenance of L&L Marketplace: a new retail outlet in Nashville.

Activities and Interests

Music Album

Released August 2021

- Wrote, recorded, co-produced, self-funded, and released a 9-track album of original music.
- Collaborated with ten creatives over six months to complete this project mid-pandemic.
- Became intimately familiar with the music production "order of operations," which takes a song from an idea to a collection of recordings and finally to a finished commercial product.

Athletics

Ongoing

- Boulderling - Working toward lifetime goal of climbing 10 V10 boulders.
- Running - Recently completed a sub 2-hour half-marathon.

Study Abroad

Summer 2023

- Studied Japanese Fiction while living in Tokyo, Japan during the summer of 2023.